ASTRA

Sovereign · Specific · Federated · Mitigated · Operational · Auditable

- Aerial Surveillance & Threat Recognition Assistant.
- Round 1: end-to-end on a 6 GB laptop GPU. Round 2: operational layer.
- Round 3: command-center polish + 22 features stacked on top.

22 features on top of R2

Same 6 GB laptop GPU. No cloud.

- Command Center wall · 12-tile live ops view
- Sidebar v2 with real-time status pills + theme toggle
- PGD adversarial training (real per-batch FGSM)
- RAG commander Q&A (Phi-3.5 + retrieval, rule-based fallback)
- JWT auth + role gates · Onboarding tour · CSV/JSON exports
- Map heatmap · Mission compare · Replay scrub · Differential privacy
- GitHub Actions CI · full-stack docker compose · Playwright backup video

Stage-2 cascade — fine-grained airframe ID

Stage-1 says 'aircraft.' Stage-2 says which one.

- MAR20 dataset · 20 military airframes · 89.03% top-1 · 99.32% top-5
- Drag in 02-airfield-hero.jpg → 14 dets in <2 s
- Cascade IDs: 6× P-3C · 2× E-8 · 2× E-3 · 2× C-130 · 1× FA-18
- Stage-2 model size: 3.2 MB · inference: 0.5 ms/crop

Real-time live push

Pins POP onto the map as the pipeline runs.

- WebSocket /ws/live · single Postgres write triggers fan-out
- LIVE pulse indicator on every connected operator console
- 5s polling fallback if the socket drops — never miss a pin

Federated learning + FoolsGold + DP

3 bases. 0 data shared. 1 better model.

- FedAvg orchestrator + FoolsGold cosine-sim Sybil quarantine
- Faithful re-impl of Fung et al. RAID 2020
- Demo: 3 honest non-IID clients + 2 colluding label-flippers
- Without FoolsGold: 0% accuracy under attack
- With FoolsGold: attackers w=0.000 quarantined, honest 76% acc
- Δ +76 percentage points from the defense alone
- R3: optional DP-SGD per-client gradient noise (Abadi 2016)

Adversarial-aware A/B

We measured the weakness AND the mitigation curve.

- Same image. Same ε. Two model variants.
- v1 baseline @ ε=0.05 (helipad): 0% retention (4 → 0 dets)
- v1-hardened @ ε=0.05 (helipad): 80% retention (5 → 4 dets)
- Δ +80 percentage points on the canonical demo image.
- R3: PGD adversarial training (per-batch FGSM, K=3)
- Honest disclosure: hardened variant is partial — full PGD is the v3+

Phone buzzes

Commander's phone, not commander's screen.

- Telegram bot integration · httpx · threshold-gated ≥ 0.85
- Markdown alert with airframe + distance + narrator text
- Operator drops ASTRA_BOT_TOKEN + CHAT_ID in .env. That's it.

Audit · Diff · Dual-control · Q&A

*What changed in *how* operators use it.*

- SHA-256 hash-chained audit · /audit/verify endpoint
- Mission Diff — added/removed/moved tracks across passes
- Dual-operator engagement — 2 distinct ops to engage; 409 on dup
- Commander Q&A — Phi-3.5 retrieval + rule-based fallback
- JWT auth · operator/commander/admin roles

5-layer sovereign stack

Every layer ships. Nothing on a 3rd-party server.

- [01] UI · React + Vite + Tailwind + Leaflet + Recharts · 9 pages
- [02] API · FastAPI + Pydantic · 20+ endpoints · /ws/live
- [03] Pipeline · YOLO → cascade → OOD → narrator → Telegram
- [04] Persistence · Postgres 16 + Redis 7 · SHA-256 chain
- [05] Federated trust · FedAvg + FoolsGold + DP-SGD

All in one chart

Every number reproducible by a script in this repo.

- Stage-1 mAP@0.5 (DOTA, 8 classes): 0.515 · aircraft 0.92 · ship 0.77
- Stage-2 cascade top-1 (MAR20, 20 airframes): 89.03% · top-5: 99.32%
- GPU latency: 22 ms/frame · CPU ONNX: 36 ms · cascade overhead: +0.5 ms
- Adversarial v1 vs v1-hardened (helipad ε=0.05): 0% $\rightarrow$ 80%
- Federated under attack (without/with FoolsGold): 0% $\rightarrow$ 76% accuracy
- Audit chain integrity: SHA-256, 100+ rows verified ok
- Total model size on disk: 8.7 MB

Sovereign + auditable + small + free

We don't claim Palantir parity. We claim the right trade.

- Sovereign on a laptop: ASTRA ✓ Palantir ✗ OSS ■ (training only)
- Fine-grained military ID: ASTRA ✓ Palantir part. OSS ✗
- Federated + Sybil resistance: ASTRA ✓ Palantir ? OSS ✗
- Adversarial-aware A/B: ASTRA ✓ Palantir ? OSS ✗
- Tamper-evident audit: ASTRA ✓ Palantir part. OSS ✗
- Cost ongoing: ASTRA ■0 Palantir ■L/mo
- Time to deploy: ASTRA 5d Palantir mos.

Trophy material.

Three publishable contributions. One project. Five days.

- Sovereign — runs end-to-end on a 6 GB laptop GPU.
- Specific — 20 fine-grained military airframes via cascade.
- Federated — 3 bases, FoolsGold quarantine, optional DP.
- Mitigated — adversarial A/B with measured retention curve.
- Operational — dual-control, mobile alerts, mission diff, Q&A.
- Auditable — SHA-256 chain · verify-on-demand.
- Ready — `python scripts/e2e_regression.py` returns ALL GREEN.